concepts Include

- Inheritance
- Method Hiding
- Method Overriding
- Polymorphism
- Sealed Classes

Each assignment will include:

- 🎯 Scenario
- 🧱 Classes to design
- 📌 Tasks to Complete

Assignment 1: Healthcare System

Concept Focus

Inheritance + Method Overriding + Polymorphism

🎯 Scenario

A hospital system manages different types of staff such as doctors and nurses. Each staff member calculates their salary differently.

🧱 Class Structure

Base Class

Staff

Derived Classes

Doctor

Nurse

LabTechnician

📌 **Tasks**

1. Create a base class **Staff**
    - Properties
        - StaffId
        - Name
        - BaseSalary
2. Create a method

CalculateSalary()

3. Create derived classes

Doctor

Nurse

LabTechnician

4. Override the method **CalculateSalary()**

Example logic:

Doctor

BaseSalary + ConsultationFee

Nurse

BaseSalary + NightShiftAllowance

LabTechnician

BaseSalary + EquipmentAllowance

Observe **runtime polymorphism**.

Assignment 2: Banking System

Concept Focus

Inheritance + Method Hiding

🎯 **Scenario**

A bank system calculates interest for different account types.

🧱 **Class Structure**

Base Class

Account

Derived Classes

SavingsAccount

CurrentAccount

📌 **Tasks**

1. Create base class **Account**

Properties

AccountNumber

Balance

Method

CalculateInterest()

Print:

Base account interest calculation

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2. Create derived classes

SavingsAccount

CurrentAccount

Hide the base method using

new keyword

Example

public new void CalculateInterest()

📌 **Expected Behavior**

Account acc = new SavingsAccount();

acc.CalculateInterest();

Observe that **base method executes**.

Explain **method hiding behavior**.

Assignment 3: E-Commerce Order System

Concept Focus

Method Overriding + Runtime Polymorphism

🎯 **Scenario**

An e-commerce system calculates shipping cost for different delivery types.

🧱 **Class Structure**

Base Class

Order

Derived Classes

StandardOrder

ExpressOrder

InternationalOrder

📌 **Tasks**

1. Create base class **Order**

Properties

OrderId

OrderAmount

Virtual method

CalculateShippingCost()

Default shipping cost:

50

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2. Override method in derived classes

StandardOrder

Shipping = 50

ExpressOrder

Shipping = 100

InternationalOrder

Shipping = 500

📌 **Test Polymorphism**

List<Order> orders

Add

StandardOrder

ExpressOrder

InternationalOrder

Loop and call

CalculateShippingCost()

Assignment 4: Vehicle Management System

Concept Focus

Sealed Class

🎯 Scenario

A vehicle system where certain vehicle types cannot be inherited further.

🧱 Class Structure

Base Class

Vehicle

Derived Classes

Car

Bike

ElectricCar

📌 Tasks

1. Create base class **Vehicle**

Properties

VehicleNumber

Brand

Method

StartVehicle()

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2. Create derived class **Car**

Add property

FuelType

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3. Create class **ElectricCar**

Inherit from

Car

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4. Declare ElectricCar as **sealed**

public sealed class ElectricCar : Car

📌 **Additional Task**

Try creating a class

Tesla : ElectricCar

Observe the **compile-time error**.

Explain why **sealed classes prevent inheritance**.

Assignment 5: Education System

Concept Focus

Inheritance + Overriding + Polymorphism

🎯 **Scenario**

An education system where different types of students have different grading policies.

🧱 **Class Structure**

Base Class

Student

Derived Classes

SchoolStudent

CollegeStudent

OnlineStudent

📌 **Tasks**

1. Create base class **Student**

Properties

StudentId

Name

Marks

Virtual Method

CalculateGrade()

Default grade rule:

Marks > 50 = Pass

Else Fail

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2. Override grading logic

SchoolStudent

Pass if Marks > 40

CollegeStudent

Pass if Marks > 50

OnlineStudent

Pass if Marks > 60

📌 **Test using Polymorphism**

Student[] students

Store objects of all student types.

Call

CalculateGrade()

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all assignments to include:

✓ Constructors
✓ Properties

Assignment 6

Create a class called furniture which has Data Members like

- OrderId
- OrderDate
- FurnitureType
- Qty
- TotalAmt
- PaymentMode (credit/debit card)

Furniture Type can be Chair/Cot

- Derive the Chair and Cot class from Furniture class

If Chair Class then data members are

Chair type – Wood/Steel/Plastic

- purpose – Home/office
- If Wood Chair then Wood Type would be Teak Wood/Rose Wood
- If Steel Chair then it would be Gray Steel/Green Steel/Brown Steel
- If Plastic Color would be Green/Red/Blue/White
- Rate

If Cot class then Data Members are

- Cot Type – Wood / Steel

Same Data Members defined for Wood/Steel chair will be applicable for this.

- Capacity – Single/Double
- Rate

All the classes should have GetData() to accept user input and ShowData() to display the details .

Create a Console Application to arrive at the Solution.

- Hint : Implement the solution using Inheritance Concept.